Tandem drum rollers operated in the static mode are the most common finish compactors. Remember, the narrower the drum width, the higher the static force exerted. It is common to see smaller compactors used in the finish position.

Vibration should not be used in the finish phase. If additional density is required, the problem needs to be addressed in the initial and intermediate phases. The goal of the finish phase is smoothness, not density gains. The finish compactor should make long, slow passes to make the mat smoother. If the finish phase on a portion of the mat is completed and the finish compactor needs to park and wait, park on only a portion of the mat or an adjoining surface that is cool enough to support the machine without deforming the mat.



Tandem drum rollers operated in the static mode are common during the finish compaction phase.

[TEST STRIPS]

On many projects, the public works department in charge will require that a test strip be successfully completed before the start of full production. The requirements for test strips vary greatly from one location to another. In general, the test strip verifies that asphalt production meets the job mix formula, that the paving equipment lays down a satisfactory mat, and that the selected compaction equipment achieves the specified density.

A test strip can be a separate element or part of the paving project. It will consist of adequate tonnage to complete the required tests. In this manual, the only concern is about the density testing.

To complete the test strip, the quality control technician or supervisor will need to have an accurate temperature measuring device and a calibrated density testing gauge. The compaction equipment used on the test strip must be the equipment that is planned for the project.

The quality control technician or supervisor should have planned a rolling pattern for each phase of compaction, and should have selected vibratory amplitude, vibratory frequency and working speed for the initial phase compactor. The vibratory characteristics are determined by type of mix and thickness of the asphalt layer and the working speed of the paver.

As soon as the paver pulls off the starting joint, begin to collect data. Once the paver has reached planned paving speed and the correct mat thickness has been established, check the temperature of the mat directly behind the paver screed. Throughout the length of the test strip, continue to check and document mat temperature. Consistent mat temperature is one of the keys to creating consistent mat density. If the mix used on the project has a tender zone, you will be able to document the temperature for the start of the tender zone and the temperature when tenderness disappears.



Check the density of the mat laid by the screed before the compaction process begins. Knowing the screed laid density will help you select amplitude and the number of passes that may be necessary to get the required density.

Check screed laid density at several places across the width of the mat. Check the density of the mat after each pass made by the initial phase compactor.



Sometimes, checking the density of the mat before initial compaction (top) is overlooked on the test strip. Density values and temperatures written on the mat (above) help compactor operators.

User Tip: *When you place the density testing gauge on the mat to make the first density check, draw an outline around the base of the gauge with chalk. The chalk outline will help you put the gauge in exactly the same spot after each pass.*

Continue to make passes with the initial phase compactor until you have reached the target density for the initial phase. Let's look at an example of a test strip created on a highway project.

[DENSITY SPECIFICATIONS]

Minimum density:	92% theoretical maximum density
Target density:	93.5 to 95.5 theoretical maximum density

[PROJECT CONDITIONS]

Asphalt mix:	25 mm (1") dense graded
Asphalt cement:	5.8% polymer-modified oil
Paving thickness:	80 mm (3.1")
Paving width:	3.66 m (12')
Screed laid density:	80% theoretical maximum density
Mat temperature:	149° C (300° F)

[VIBRATORY CHARACTERISTICS, INITIAL PHASE]

Drum width:	200 cm (79")
Static force per drum:	29.7 kg/cm (166 lb/in)
Amplitude:	0.78 mm (0.031")
Frequency:	42 Hz (2520 vibrations per minute)

[TEST STRIP RESULTS, INITIAL PHASE]

Pass One:	84%
Pass Two:	87%
Pass Three:	90%
Pass Four:	92%

[VIBRATORY CHARACTERISTICS, INTERMEDIATE PHASE]

Drum width:	200 cm (79")
Static force per drum:	29.7 kg/cm (166 lb/in)
Amplitude:	0.30 mm (0.012")
Frequency:	63.3 Hz (3800 vpm)

[TEST STRIP RESULTS, INTERMEDIATE PHASE]

Pass Five:	93%
Pass Six:	94%

[STATIC CHARACTERISTICS, FINISH PHASE]

Drum width:	170 cm (67")
Static force per drum:	31.8 kg/cm (178 lb/in)

[TEST STRIP RESULTS, FINISH PHASE]

Pass Seven:	94.5%
Pass Eight:	95.0%

Summary: *This test strip required four passes in the initial phase to reach the minimum density acceptance level. Two more intermediate passes added enough density to meet the target specification. Finish compaction added another percent. The compaction process met the requirement for density, but did it meet the production requirement? We can use the Cat Interactive Production Calculator to confirm that compaction production matches the paving production.*

PAVER SPEED CALCULATOR

- Trucking
- **Paver Speed**
- Compaction
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

- Exit

General Inputs

Paving Thickness:	[3.15] *in*	[80.0] *mm*
Paving Width:	[12.00] *feet*	[3.658] *meter*
Material Density Uncompacted:	[127] *lbs/ft³*	[2034] *kg/m³*

Paver Speed @ Given Production Rate

Production Rate of Hot Plant:	[220] *tons/hr*	[200] *tonnes/hr*
Calculated Paving Speed - 100% Efficiency:	[18.3] *ft/min*	[5.58] *m/min*
Calculated Paving Speed - 95% Efficiency:	[19.2] *ft/min*	[5.86] *m/min*
Calculated Paving Speed - 90% Efficiency:	[20.1] *ft/min*	[6.14] *m/min*
Calculated Paving Speed - 85% Efficiency:	[21.0] *ft/min*	[6.42] *m/min*
Calculated Paving Speed - 80% Efficiency:	[22.0] *ft/min*	[6.70] *m/min*
Calculated Paving Speed - 75% Efficiency:	[22.9] *ft/min*	[6.97] *m/min*

Effective Paving Speed:	[18.3] *ft/min*	[5.58] *m/min*

The production rate planned for this project is 200 tonnes per hour (220 tons per hour). You have checked the screed laid density and can, therefore, accurately calculate the weight of the material laid by the screed. On this project, screed laid weight is 2034 kg/m³ (127 lb/ft³). The paving depth is 80 mm (3.15") and the paving width is 3.66 m (12'). The effective paving speed is therefore 5.58 meters per minute (18.3' per minute). You now have to confirm that the initial phase compactor can keep up with the paver.

COMPACTION CALCULATOR

- Trucking
- Paver Speed
- **Compaction**
- Windrow
- Yield
- Slope
- Thickness
- Job Summary
- Legal

- Exit

Roller Model: [*Click to Select Another Model*] [**CB54 XW**]

General Inputs

Paving Width:	[12.00] *feet*	[3.658] *meter*
Actual Drum Width:	[79] *in*	[200.66] *cm*
Amount of Overlap:	[6.0] *in*	[15.2] *cm*
Speed of Vibrator:	[2520] *VPM*	[2520] *VPM*
Impacts (recommended):	[12] *per ft*	[38] *per m*
Number of Passes to Cover Mat Width Once:		[2]
Number of Repeated Passes (from test strips):		[4]
Total Number of Passes:		[9]
Roller Efficiency Rate (recommended 75 to 85%):		[80] *%*

			Effective Paver Speed
Actual Roller Speed:	[210] *fpm*	[64] *mpm*	[18.3] *ft/min*
Effective Roller Speed*:	[19] *fpm*	[6] *mpm*	[5.58] *m/min*

** Effective Roller Speed should be at least 100% but no more than 115% of the Effective Paver Speed.* [**% = 104**]

The compactor selected for the initial phase has 200 cm (79") wide drums. This compactor can cover the mat in two overlapping passes. On the test strip, four passes were needed to reach minimum acceptance density. The calculator shows that this will be a nine-pass pattern. At an actual working speed of 64 meters per minute (210 feet per minute), the initial phase compactor will match the paving speed. The vibratory frequency has been set at 42 Hz (2520 vibrations per minute). Frequency and working speed in this example combine to create 38 impacts per meter (12 impacts per foot). That impact spacing is ideal for consistent density and smoothness.

The test strip has been successfully completed from the standpoint of density measurements taken in the field. Cores extracted from the asphalt layer and analyzed in a laboratory will confirm the densities and allow you to further calibrate the density testing gauge, if necessary.



Cutting cores and lab analysis provide the final confirmation that the test strip test density meets the target.

[GETTING MORE DENSITY ON THE TEST STRIP]

If the initial phase compactor using the selected vibratory settings and rolling pattern does not achieve the required density, changes will be necessary. Sometimes you can continue working on the same area of the test strip if the mat is still hot enough. Following is a list of ways to increase density:

- **Add more passes.** You can add more passes only if you can continue to match the paving speed.
- **Increase amplitude.** If a higher amplitude setting is available and the asphalt layer will accept the additional force without causing drum bounce, select the higher amplitude.
- **Increase tire pressure or ballast weight.** If a pneumatic compactor is used for initial compaction, you should be able to increase the compaction force without damaging the mat.
- **Use a higher production compactor.** If one is available, a compactor with wider drums or higher amplitude may be substituted. A compactor with wider drums may cover the mat in fewer overlapping passes and create a faster pattern that makes it easier to add passes.
- **Work closer to the paver.** By keeping the initial compactor very close to the paver, you will be working on the hottest asphalt. You may have to shorten the length of the rolling pattern to accomplish this.
- **Slow the working speed of the compactor.** A slower working speed delivers more force to the area being covered because the impact spacing is closer. This assumes that the slower speed will still allow the compactor to keep up with the paver.

Summary: *Setting up the compaction process and successfully completing test strips require planning and gathering data. Assuming a process that was successful on one project will also be successful on another project is very risky. True, experience is a good teacher, but there are many variables that affect density. Each project deserves a full analysis prior to starting work.*


CB534D
CAT



Unit 5
ROLLING PATTERNS

By using the Cat® Interactive Production Calculator and creating cooling curves, you can plan successful patterns prior to the start of your project.

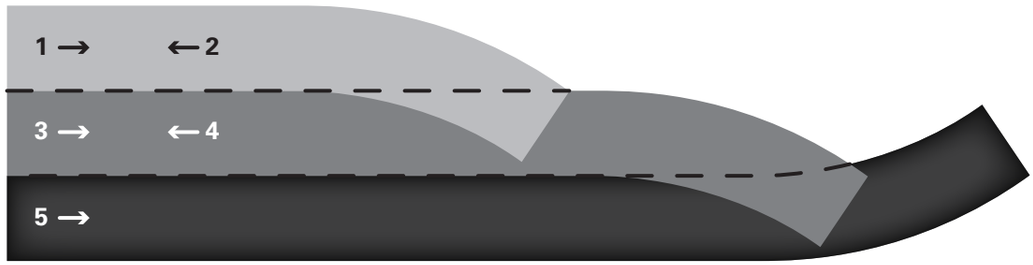
A rolling pattern is a series of movements made by a compactor or compactors on a freshly-laid, uncompacted asphalt layer. The rolling pattern is intended to be consistently repeated in order to produce uniform density in the asphalt layer.

The rolling pattern covers a certain area of square meters (feet) defined by the length and width of the pattern. It is assumed that the thickness of the asphalt layer is relatively consistent from one edge of the mat to the other within the pattern. The temperature of the asphalt mix within the rolling pattern will also be reasonably consistent as long as the area covered by the pattern is always in the same relationship to the paver as the paver moves forward. Therefore, a rolling pattern with a consistent number of passes, a consistent working speed and consistent compaction forces should produce uniform density.

Note: In this manual, the term "pass" shall mean the movement of the compactor in one direction. In other words, when the compactor begins a pattern by moving forward from a starting point to a point closer to the paver, that movement is considered one pass. When the compactor reverses to return to the starting point of the pattern, that movement is considered another pass.

Once a rolling pattern has been established, it should not be altered unless there are changes in the paving process in front of the compactor, changes in the mix formula or changes in climate conditions.

REVERSING



[BASIC ROLLING PATTERN]

There are certain techniques that are common to any rolling pattern. One technique is stopping and reversing a tandem drum roller at the end of a pass.

In the drawing shown above, it is assumed that the mat has two unconfined edges or that there is no adjacent cold mat to roll onto. The compactor operator must stop and reverse on the hot mat.

Notice that the first two passes are along one edge of the mat. At the end of Pass One, the operator turns toward the center of the mat and stops slowly with both drums turned at least 30° leaving the stop mark at an angle to the direction of compaction. The operator reverses in the same path for Pass Two.

Pass Three is down the center of the mat with some overlap on the coverage of Passes One and Two. Pass Three is longer than Pass One in order to keep up with the paver and to clean up the stop mark left at the end of Pass One.



Two initial phase compactors reversing direction behind a paver. Note angled stop.

At the end of Pass Three, the operator turns toward the uncompacted edge, being careful to not push out the edge of the mat. Again, the stop mark is left at an angle to the direction of compaction. The operator reverses in the same path for Pass Four.

Pass Five is along the other unconfined edge with some overlap on the coverage of Passes Three and Four. Pass Five continues through the stop mark left at the end of Pass Three. At the end of Pass Five, the operator turns toward the center of the mat leaving an angled stop mark where the next pattern will clean it up. The operator reverses in the same path for Pass Six.

Pass Seven will reposition the compactor to start another pattern. This is called a seven-pass pattern.

This pattern results when it takes three overlapping passes to cover the width of the mat and it takes two passes per coverage to create the required density.

Initial phase compactors are always stopping to reverse in proximity to the rear of the paver. There are no absolute rules that dictate how far behind the paver the compactor should stop. Workplace safety should be the primary consideration. A reasonable guideline would be for the compactor(s) to stop at least 5 m (16') behind the paver. Remember, there may be laborers or screed operators working on the mat behind the paver.

User Tip: *When stopping a steel drum roller to reverse direction, whether on a hot mat or a cold adjacent mat, always shut off the vibratory system as soon as you begin to slow down. Remember, it is important to maintain drum impact spacing. As the machine speed decreases, the impacts may be too close together. You can manually deactivate the vibratory system or you can select the "AutoVibe" feature that will automatically stop and start the vibratory system when the working speed reaches programmed levels.*



Pneumatic compactors are allowed to stop straight on the mat.

Unlike steel drums, the rubber tires on pneumatic compactors should not be turned when bringing the compactor to a stop. Aggressive turns with pneumatic compactors will tear the mat. The pneumatic compactor should stop slowly without turning. There will be a slight stop mark in the mat, but usually the finish roller will clean up those marks completely.

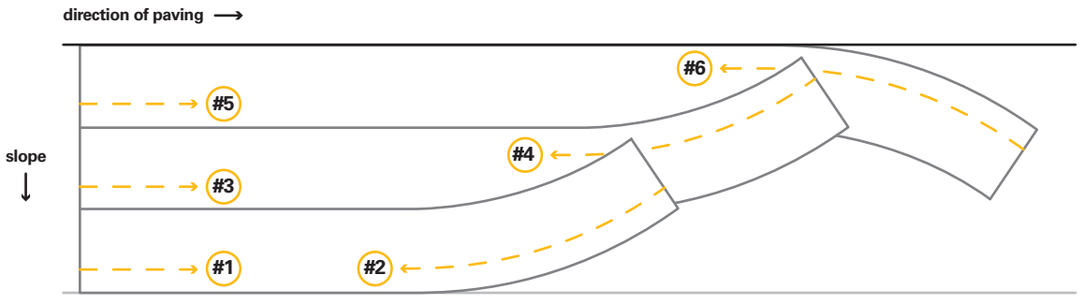
[PATTERN FOR TWO UNCONFINED EDGES]

In this example, assume that the mat has two unconfined edges, that the left edge is the centerline of the structure and that there is a 2% slope from the centerline to the right unconfined edge.

When the structure to be compacted has two unconfined edges and a sloped surface, Caterpillar recommends making the first passes along the lower edge of the structure. The next series should be in the center of the mat. The final passes should be along the upper unconfined edge. Compaction from the low side to the high side tends to build strength in the mat and reduce the amount of mat deformation.

In general, the First Pass along any unconfined edge should be made with the drum edge at least 15 cm (6") away from the edge. The Second Pass, typically the return pass in the same coverage area as the first pass, should be made with the drum slightly overlapping the edge. This sequence also helps minimize mat distortion.

TWO UNCONFINED EDGES



When compacting unconfined edges, watch for cracks in the mat along the drum edge when the drum edge is set back from the unconfined edge. Some mixes with large aggregates and low asphalt cement content will show deep cracks if the edge is not overlapped during the first pass.

When cracks appear, immediately change the rolling pattern to overlap the unconfined edge with every pass along the edge of the mat.

Pneumatic compactors should not overlap unconfined edges. The rubber tires should be at least 15 cm (6") away from the unconfined edge to avoid rolling over or distorting the edge of the mat.



Crack in the mat along drum edge set back from the unconfined edge.



Pneumatic compactors always stay al least 15 cm (6") from unconfined edges.

[PATTERN FOR ONE CONFINED EDGE]

In this example, assume that the left edge of the mat is matching an adjacent mat along the centerline of the structure. The adjacent mat is compacted and is cold. Traffic cones are set adjacent to the centerline edge on the cold mat and traffic is present on the cold mat. A 2% slope runs from the centerline down to the unconfined edge. You have two acceptable rolling patterns for this application.

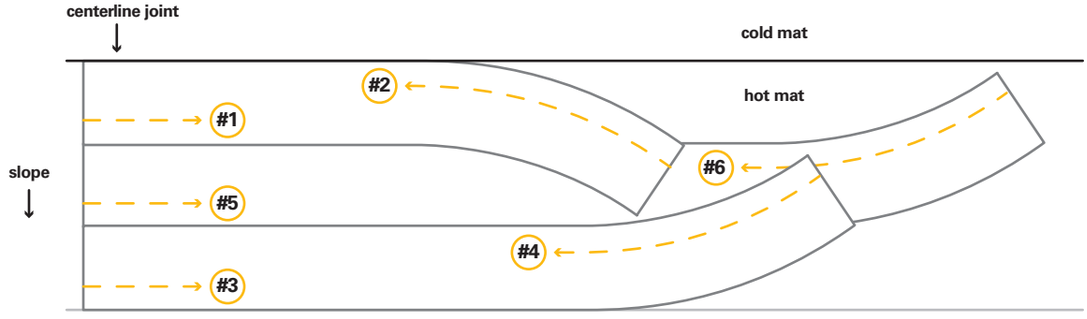
If there is a joint density specification on the project, then the first pass should be along the left edge of the mat to take advantage of the highest mat temperature resulting in the highest joint density. Both drums should be completely on the hot mat about 15-30 cm (6-12") away from the cold mat. During Pass Two, the return pass along the left edge, the drums should be positioned to overlap the hot / cold joint by about 15 cm (6"). The overlap will begin the process of sealing the longitudinal centerline joint.

Passes Three and Four will be along the unconfined edge to build strength and minimize mat deformation at the unconfined edge.

Passes Five and Six are made in the center of the mat. This portion of the mat will be the coolest by this time, but the center portion of the mat will have, in effect, two confined edges to aid in compaction.

If there is no joint density specification, then Passes One and Two can be along the right edge on the low side of the sloped mat as shown in the illustration for two unconfined edges. The center of the mat is compacted by Passes Three and Four. Finally, the centerline joint is compacted with Pass Five slightly off the joint and Pass Six overlapping the joint.

ONE UNCONFINED EDGE





Pneumatic compactors in the intermediate phase are effective at sealing hot / cold longitudinal joints.



There are no drum stop marks in the fresh mat when the compactor is able to roll onto a cold, compacted mat.

All other compactors during intermediate or finish phases may overlap the longitudinal joint. The rubber tires of pneumatic compactors are particularly efficient at "pinching" joints. The pneumatic compactor operator should try to straddle the joint with one of the tires.

On some projects, the edge of the mat that is matching a joint will be adjacent to a cold, compacted mat. Whenever possible, compactor operators should roll off the hot mat and onto the cold mat to stop and reverse direction. By reversing direction on the cold mat, there will be no stop marks left on the hot asphalt layer and smoothness will be improved.

The compactor operator must be aware of several safety issues when rolling off the hot mat to stop and reverse. First, there may be traffic using the adjacent lane. Pilot vehicles may be leading traffic through the work zone. The operator should never pull onto the adjacent mat if there is traffic present.

Second, there may be workers around the paver. In particular, laborers may be raking the joint behind the paver. Be sure to pull off the mat far enough behind the paver if there are workers present.

[PATTERN USING EMERGENCY LANE FOR REVERSING]

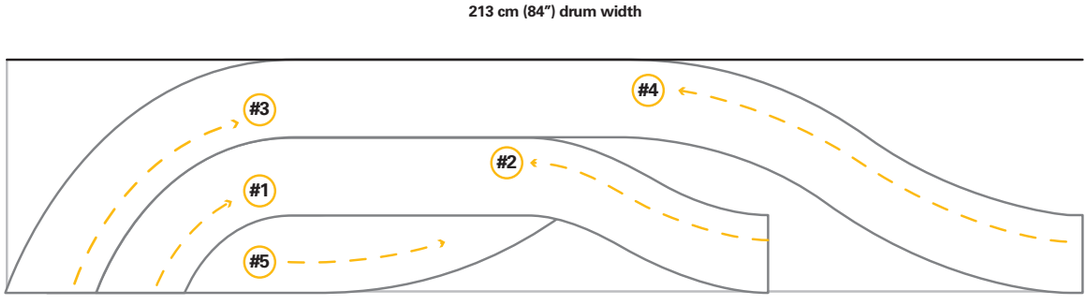
On some projects, the plan calls for the paving width to include an emergency lane (also referred to as a shoulder or breakdown lane) along with one driving lane. Normally, if the emergency lane is less than 1.5 m (5') wide, it will be included in the conventional pattern used by the initial phase compactor. Or, if the emergency lane has a separate slope, sometimes the emergency lane will be compacted by a utility compactor and will not be included in the pattern with the driving lane.

However, if the emergency lane is at least 1.5 m (5') wide, the emergency lane can be included in the initial phase pattern and can be used for all compactor stopping and reversing.

The rolling pattern will resemble a series of half circles. After each pass, forward and backward, the compactor operator will arc slowly across the driving lane and straighten out on the emergency lane with both drums on the emergency lane. The compactor operator will stop straight. Stopping straight is generally permitted on emergency lanes since there is no smoothness specification for emergency lanes.

If a pneumatic compactor is part of the compaction train, the pneumatic compactor will continue to stop straight on the driving lane and will not turn off onto the emergency lane. The finish compactor should use the emergency lane for stopping and reversing, too.

REVERSING ON EMERGENCY LANE

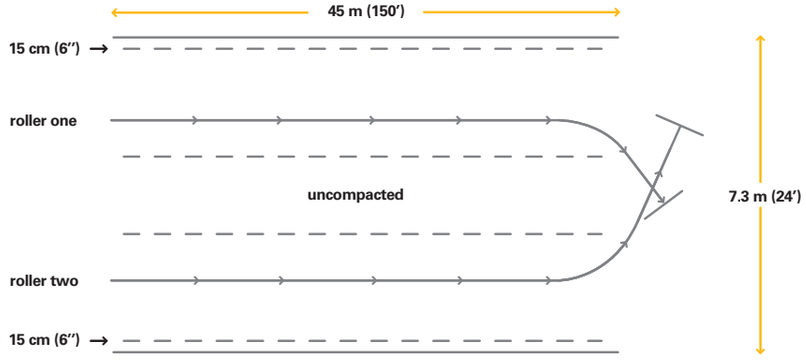


[ECHELON COMPACTION PATTERNS]

On some projects, two or more compactors may operate in the initial compaction position directly behind the paver. For the following reasons, an echelon pattern should be selected.

- **Wide width paving.** When the paving width exceeds 6 m (20'), it is unlikely that one compactor will be able to cover the paving width in three or fewer passes. Therefore, in general, one initial compactor will not be able to match the paver production.
- **Stiff mix requiring many passes.** Some mix designs, especially those including modified asphalt cement, are very stiff and require many passes to bring density up to required levels. In that situation, the pattern needed for one compactor will cause that compactor to fall behind the paver.
- **Limited time for initial compaction.** The time available for initial compaction may be limited by the thickness of the mat, ambient temperature, or the appearance of a tender zone in the mat. Sometimes, more than one initial compactor is needed to cope with rapid temperature loss and a short opportunity to get initial density.

INITIAL PHASE — PASS ONE

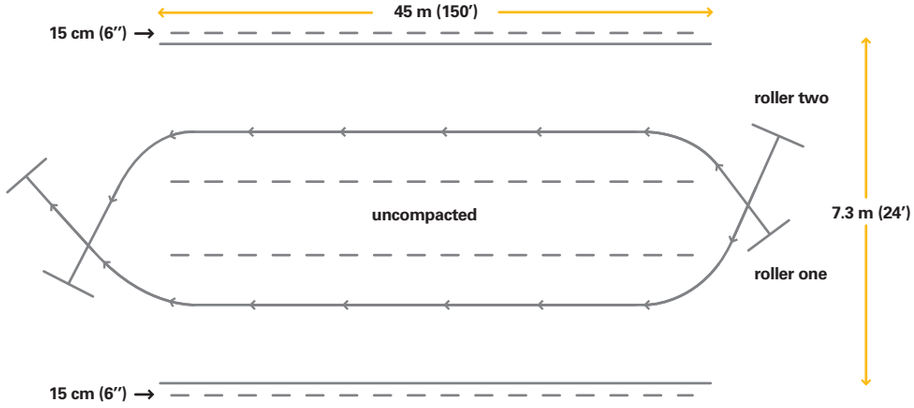


In the first example, assume that the paver is laying down 275 tonnes per hour (300 tons per hour) at a width of 7.3 m (24') and at a depth of 50 mm (2"). The effective paving speed is 6 meters per minute (20 feet per minute). There are two 200 cm (79") wide tandem drum rollers available for initial compaction. It takes two passes to bring density to the compaction target of the initial phase.

Roller One starts first along the left edge with the outer drum edge about 15 cm (6") away from the unconfined edge. Roller Two begins just after Roller One and operates along the right edge, also staying away from the unconfined edge. Roller One comes to a slow stop at an angle in the center of the mat and reverses. Roller Two moves slightly past the Roller One stop and also turns toward the center and reverses.

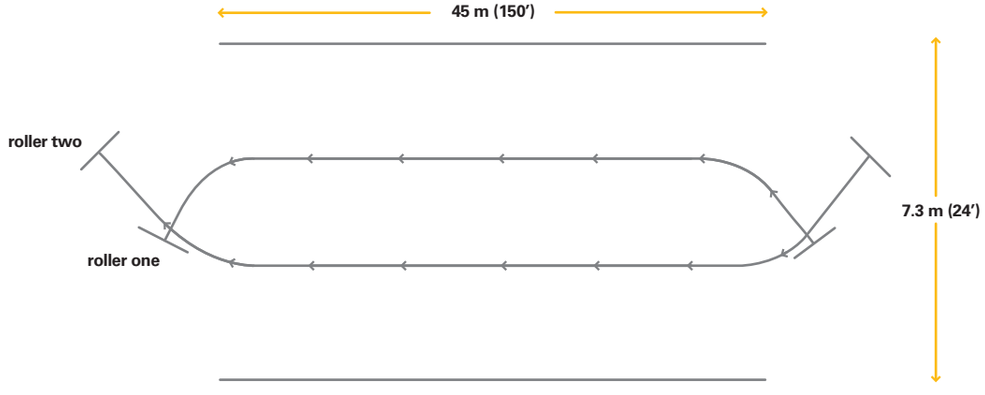
User Tip: *When employing an echelon pattern, the lead roller should be far enough in front of the second roller that the operator can complete the stop and reverse maneuver before the second roller starts to turn and reverse.*

INITIAL PHASE — PASS TWO



During Pass Two, the return to the starting point, Roller One is in the lead with Roller Two slightly behind. During Pass Two, the outer drums slightly overlap the unconfined edges. Again, both compactors turn toward the center of the mat to make the stop and reverse. At this point, the outside edges of the mat have been compacted twice. There is a strip in the center of the mat that is about 3.5 m (11.5') wide.

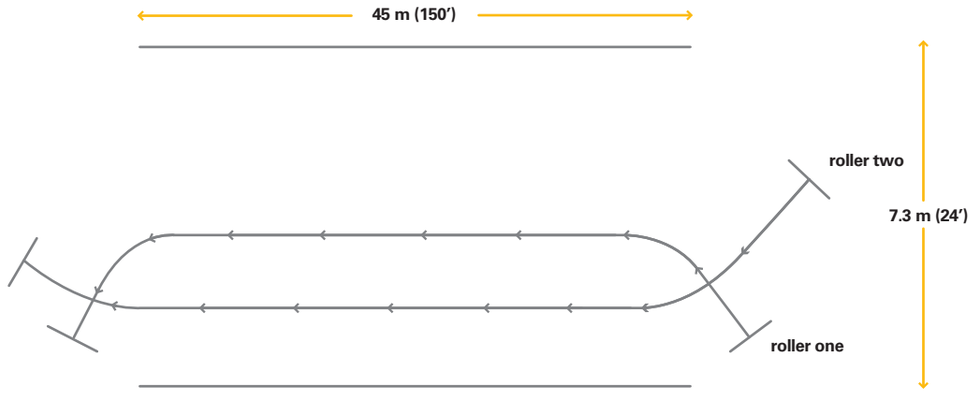
INITIAL PHASE — PASS THREE



For Pass Three, Roller One goes first and operates on the left center of the mat with a slight overlap of the left drum edge onto the area covered in the first two passes. Roller One will pass straight over the drum stop mark left by Roller Two, moving forward about 8 m (25') past the stop mark before turning toward the right edge to stop and reverse.

Roller Two is slightly behind and operates on the right center portion of the mat with the right drum edge slightly overlapping onto the area covered in the first two passes. Roller Two will clean up the first stop mark left by Roller One, continue forward another 8 m (25') past the stop mark and turn toward the left edge to stop and reverse.

INITIAL PHASE — PASS FOUR



During Pass Four, the two compactors return in the same area back to the starting point with Roller One slightly ahead of Roller Two. It is recommended that they roll straight through the stop marks they left at the end of Pass Two before stopping and reversing.



Stopping to reverse repeatedly in the same area will overwork the fresh asphalt.

User Tip: *Whenever possible, use a pattern that cleans up compactor stop marks. Do not stop and reverse in the same area. Stopping and reversing in the same area can distort the mat and create bumps that cannot be cleaned up. While it is most important to avoid stopping in the same area when the compactors move forward and reverse on the hot mat behind the paver, it is also a good practice to stagger the stop marks at the end of the return passes.*

Pass Five will be a static pass moving forward toward the paver. Both compactor operators should position their machines along the edges of the fresh mat and start the vibrator systems when the compactors enter the uncompacted zones. There should be a new pattern area about 36 m (120') long in front of the old pattern.

User Tip: *If the new pattern is too short, in other words, the paver has not moved far enough forward, the compactor operators should reduce their working speeds during Pass Five. Pass Five is done in the static mode so there is no concern about drum impact spacing. Reduce the compactor speed, but never park on the fresh mat.*

User Tip: *Sometimes the area left uncompacted in the center of the mat is relatively narrow. In that situation, there will be a large overlap between the drums of the two compactors in the center of the mat. Because a lot of the drum surface will be vibrating on an already dense mat, there is likelihood that the drums will bounce. The operators should be ready to reduce the amount of force being delivered by operating with one drum vibrating and one drum static.*



[ECHELON PATTERN WITH TWO ROLLERS]

Next, let's look at an echelon pattern using two compactors on a mat that is 4.6 m (15') wide. The paver is laying down 360 tonnes per hour (400 tons per hour) at a loose depth of 76 mm (3"). A material transfer device is feeding mix to the paver and the paving speed is 9 meters per minute (29 feet per minute). The density of the mat passing under the vibratory screed is 80% of theoretical maximum density.

On the test strip, you verified that it takes four vibratory passes at medium-high amplitude to reach target density for the initial phase. Compactor working speed is 70 meters per minute (230 feet per minute). Vibratory frequency is 42 Hz (2520 vibrations per minute). The impact spacing is therefore 36 impacts per meter (11 impacts per foot).

The temperature of the asphalt layer as it passes under the screed is consistently around 150° C (300° F). When the mat cools to around 115°C (240° F), it becomes tender. The tenderness persists until the mat has cooled to 85° C (185° F).

Start Date/Time [7/3/2011 ▾] [7:08 AM]

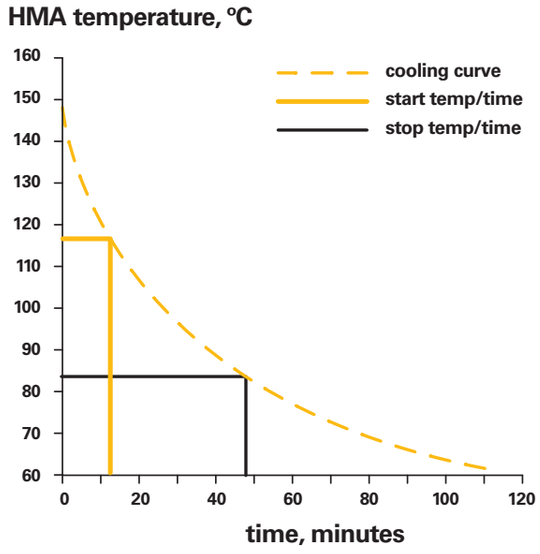
Environmental Conditions
Air Temperature: [15.6] °C
Wind Speed: [8] km/h
Sky: [Clear & Dry ▾]
Latitude: [45] °N

Mix Specifications
Mix Type: [Fine/Dense Graded ▾]
Binder Grade, PG: [64 ▾] [-28 ▾]
Lift Thickness: [76] mm
Delivery Temperature: [149] °C

Existing Surface
Material Type: [AC ▾]
Material Condition: [- ▾] [- ▾]
Surface Temperature: [15.6] °C

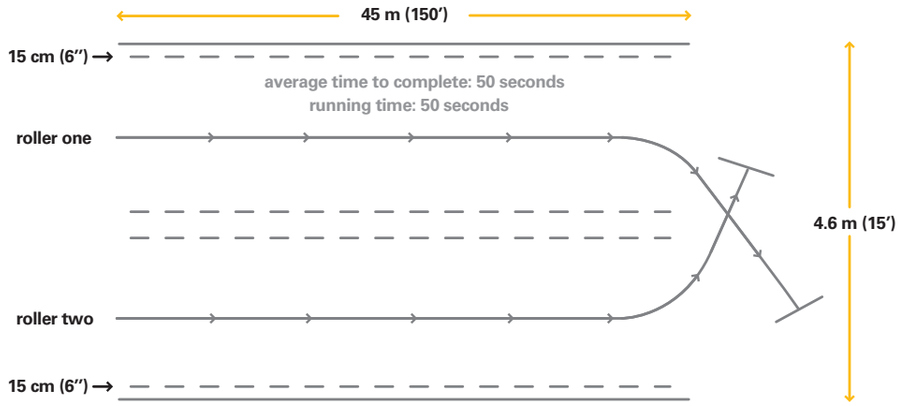
Recommended Times
Start Rolling: [13] minutes after laydown
Stop Rolling: [45] minutes after laydown

Units ◉ SI ○ English



The cooling curve shows that at the start of the shift with the ambient temperature about 16° C (61° F), there are 13 minutes available for compaction behind the paver in the initial phase before the mat becomes tender. The tenderness lasts another 32 minutes. Intermediate and finish phases of compaction can start 45 minutes behind the paver. The more important problem is to determine if the initial phase can be accomplished in less than 13 minutes using two compactors.

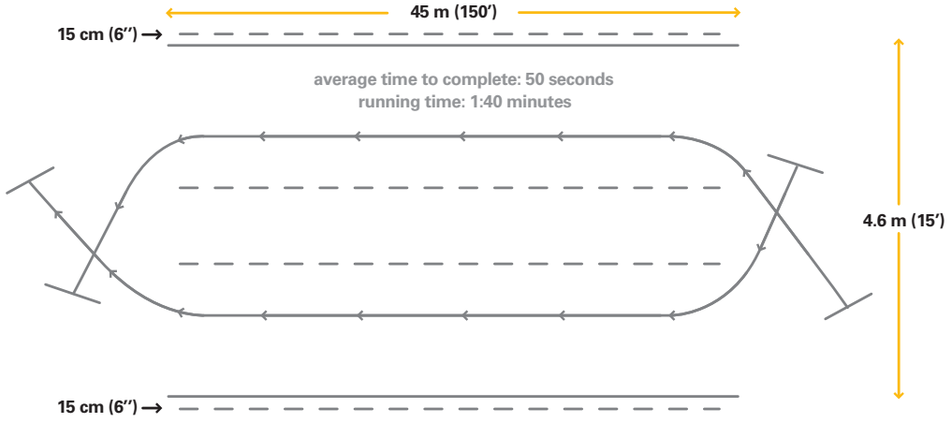
INITIAL PHASE — PASS ONE



The first pass in the initial phase of compaction is about 45 m (150') long. Roller One starts first and compacts the area along the left edge with the drum edge set back about 15 cm (6") from the edge. Roller Two is slightly behind and is compacting the area along the right side of the mat with the drum set away from the edge. Both compactors turn toward the center of the mat to stop and reverse directions. As soon as Roller Two completes its reversing maneuver, both compactors start the second pass. At a working speed of 70 meters per minute (230 feet per minute), Pass One consumes 50 seconds.

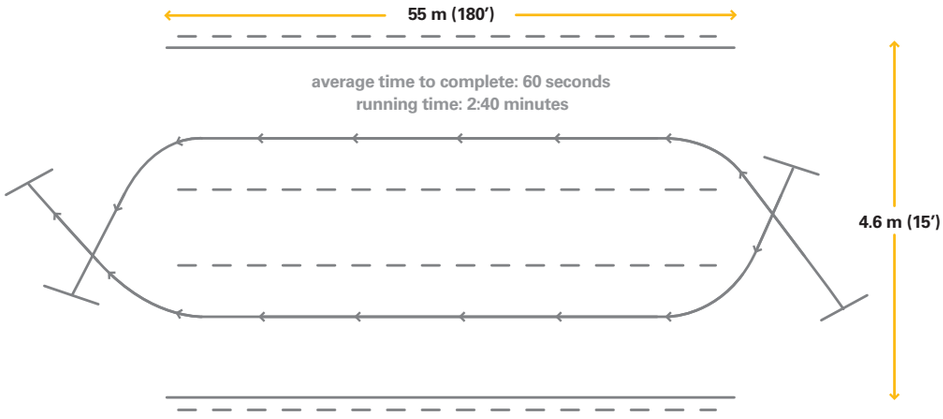
***Note:** When calculating the time it takes a compactor to complete a pass, use an efficiency factor of 75%. It accounts for the time spent slowing the compactor and coming to an angled stop prior to reversing.*

INITIAL PHASE — PASS TWO



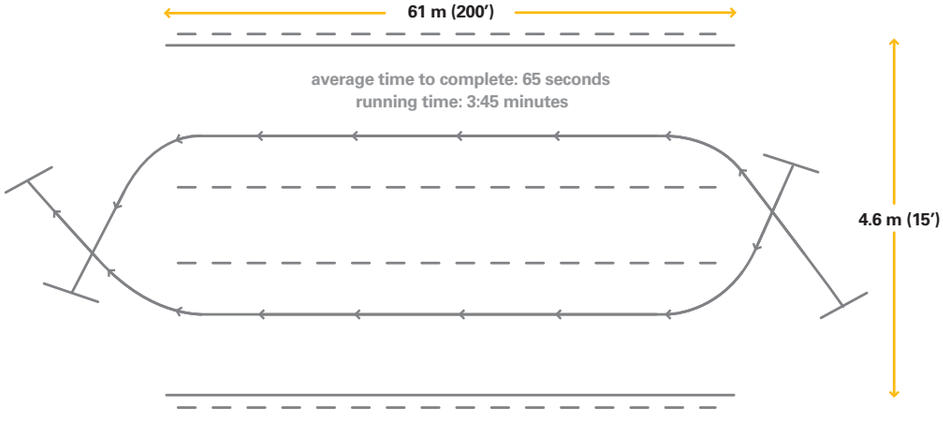
During the second pass, both compactors' drums are set to slightly overlap the edges. Roller Two is in the lead on the return pass and both rollers turn to the center to reverse direction. The second pass takes another 50 seconds. Total running time is 1 minute and 40 seconds.

INITIAL PHASE — PASS THREE



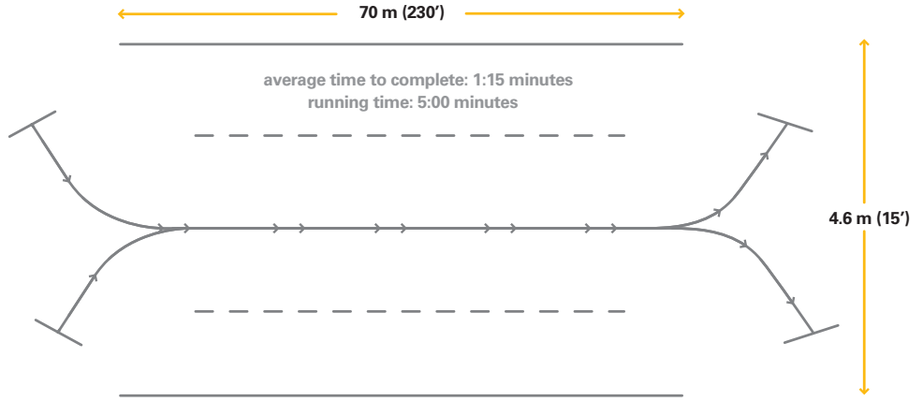
During Pass Three, Roller One goes back in the lead and extends the pattern length by another 10 m (30') in order to roll past the previous stop marks and to move the pattern closer to the paver. This pass will consume about 60 seconds and the total running time is 2 minutes and 40 seconds.

INITIAL PHASE — PASS FOUR



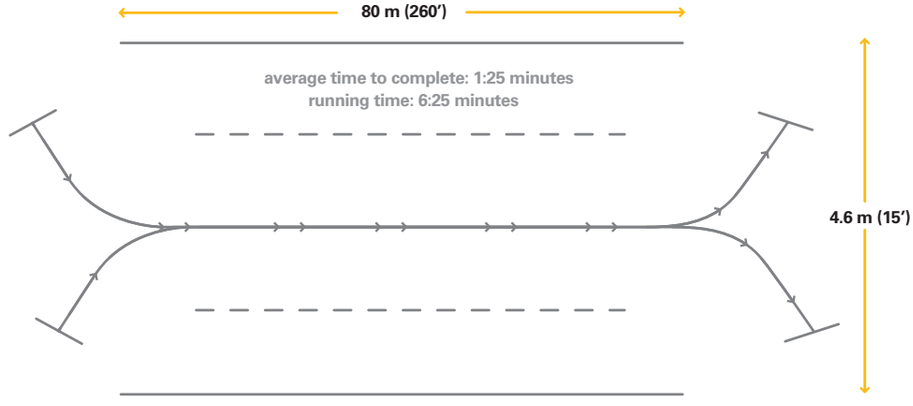
During Pass Four, both compactors return back along the edges. The pattern increases a little in length as both compactors work past the previous stop marks. Now, the right edge and left edge of the mat have been rolled four times, the number of passes verified by the test strip. The running time is at 3 minutes and 45 seconds. A strip about 1.3 m (52") wide remains in the center of the mat. Since the drum width of the two compactors is 1.7 m (67"), the two compactors can now operate in single file to complete coverage of the mat.

INITIAL PHASE — PASS FIVE



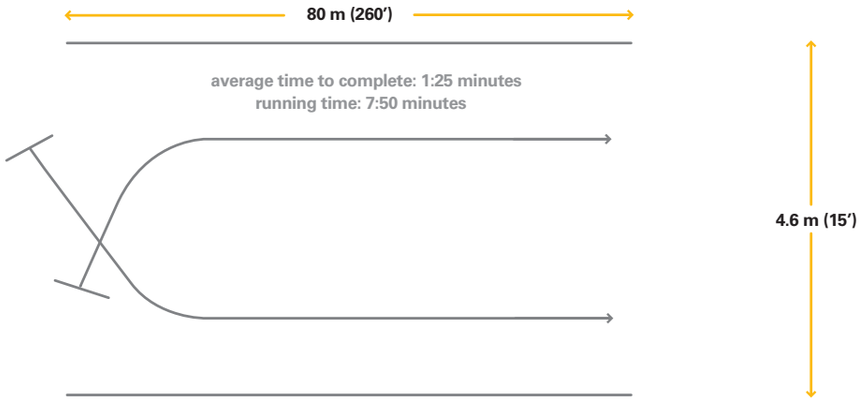
The width of the uncompacted area is slightly less than the width of the drums so there is a small overlap on both sides of the drum. Since there is a small overlap, there is less need to worry about drums bouncing on the denser portion of the mat. Roller One and Roller Two turn toward the edges of the mat at the end of the pass. This pass is again slightly longer to keep up with the paver's progress. Pass Five will take 1 minute and 15 seconds for a running time of 5 minutes at this point.

INITIAL PHASE — PASS SIX



Both compactors come back down the center of the mat in single file during Pass Six. They roll a little farther back through their stop marks. At this point, every part of the asphalt layer has had four vibratory passes. Density should be consistent across the width and length of the pattern. The running time is 6 minutes and 25 seconds.

INITIAL PHASE — PASS SEVEN



The last pass, Pass Seven, will be completed in the static mode as both compactors reposition themselves along the edges of the mat to start a new pattern. A little less than 8 minutes has been used by the two compactors assigned to the initial phase of compaction. Their pattern will stay well ahead of the tender zone.

Note: *The examples shown in this section illustrate two-compactor echelon patterns. Three-compactor echelon patterns are used less frequently but may be needed on wide-width paving applications using hard-to-compact, stiff mixes.*

Summary: *Compactor operators and quality control personnel should understand how to set up rolling patterns that accomplish three goals. First is to achieve the specified density. Second is to match the production rate of the paver. Third is to create patterns that take advantage of opportunities to ensure mat smoothness. By using the Cat Interactive Production Calculator and creating cooling curves, you can plan successful patterns prior to the start of the project.*


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